

different potato cultivars viz. Sekura, Zhukovskij Rannij, Vega and Red Scarlett with preliminary symptoms of wilt disease were sampled for *Fusarium* spp. isolation. To obtain isolates root necks of diseased plants were surface sterilized in 0.5% sodium hypochlorite for 2 minutes with successive thorough rinsing in sterile distilled water, and then transferred onto sterile Potato Dextrose Agar (PDA) and Czapek Dox Agar (CDA) supplemented with 50 mg/L chloramphenicol to inhibit bacterial growth. Pure *Fusarium* isolates were transferred onto fresh sterile PDA, with subsequent incubation at 25±2°C for 7 days under 12 hr alternating light (black/white). Obtained fungi were identified as *Fusarium* isolates based on their macro and microscopic morphology. For species identification, the DNA of the isolates was isolated by the phenol-chloroform method and used for the amplification of the 5.8S rRNA gene: conserved regions (internal transcribed spacer) using the primers ITS1 (TCCGTAGGTGAACCTGCGG), ITS4 (TCCTCCGCTTATTGATA TGC).

The pathogenicity of the isolates was assessed by their ability to cause FDR when artificially inoculated into tubers with fungal spores. Using a micro-syringe, 20 µL of fungal spore suspension (10⁵/ml) was inoculated into each of 10 tubers of the cultivars Zhukovskij Rannij, Red Scarlet and Reggi. Tubers inoculated with normal sterile saline were used as negative controls. Wounded tubers were rapped separately in sterile filter papers, placed in sterile labelled paper bags, and incubated in a humid chamber at 25°C for 21 days. After a continuous incubation of 7 days, tubers were then checked daily for external signs of dry rot disease until the 21st day of incubation. Following 21 days of incubation, the number of days until the appearance of FDR symptoms was recorded for each isolate. Tubers were then cut across the wound, the coverage of FDR was measured, and the tuber resistance was scored based on an inverse proportion of affected tuber tissue using a 9-point scale: where 9 is the absence of damage to the tuber tissue (completely resistant), and 7, 5, 3 and 1 correspond to lesions of 1–25, 25–50, 50–75 and 75–100% of tuber tissue, respectively (Bomok, 2019). Cultivar resistance indices [C_{RI}] were determined using following Equation 1.

$$CRI = \frac{\Sigma T_H - \Sigma T_A}{\Sigma T_H} \quad (\text{Eq. 1})$$

Where, T_H and T_A represent the scores of healthy and affected tubers for that particular cultivar, respectively. In this case, if $1 \geq C_{RI} > 0$, the cultivar was concluded to be resistant, while a cultivar was characterized as susceptible if $0 > C_{RI} \geq -1$.

To confirm Koch's postulates, the BOX- and ERIC-PCR methods, which are used for DNA fingerprinting in bacteria and fungi were carried out to compare the original and re-isolated pathogenic *Fusarium* spp. (Mishra *et al.*, 2015).

The scores of tuber resistance were subjected to a normal distribution test, and a Kruskal-Wallis (H) test to assess the influence of cultivar type on the manifestation of FDR using the SPSS software version 25.0, taking into account all replicates per treatment.

RESULTS AND DISCUSSION

Fungal Isolation and Molecular Identification

A total of 14 fungal isolates belonging to the genus *Fusarium* were obtained from the root necks of sampled wilting potato plants. Five isolates *Fusarium* spp. were obtained from the plants of the Sekura cultivar, in some instances, two isolates were isolated from one plant. Three isolates each were obtained from varieties Zhukovskij rannij, Vega and Red Scarlett, respectively. All sequence data exhibited between 97 to 100% homology among the tested isolates. The isolate AM9 was identified as *F. solani*, while the remaining thirteen were identified as *F. oxysporum*. All obtained sequenced data have been uploaded into the NCBI Genebank under accession numbers provided in Table 1. Similar to these results, *F. oxysporum* was confirmed as the most common potato wilt-associated pathogen in Tunisia (Ayed *et al.*, 2006). This species was ranked the leading cause of dry rot in seed potato tubers in Michigan, USA (Gachango *et al.*, 2012). Thus, being the dominant party among isolates, *F. oxysporum* could present a greater threat to potato production in the middle Volga Region.